

Advanced Calculator in C

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Language: C Programming

Purpose: This program performs basic and advanced mathematical operations such as addition, subtraction, multiplication, division, power, square root, and modulo using user input.

Features:

- Performs basic arithmetic operations: Addition, Subtraction, Multiplication, Division.
- Supports advanced functions: Power (x^y) and Square Root ($\sqrt{x}$).
- Includes Modulo operation to find the remainder between two integers.
- Error handling for invalid inputs like division by zero and negative square roots.
- Loop structure allows continuous calculations until user chooses to exit.

How to Compile and Run:

1. Save the code in a file named 'calculator.c'.
2. Open terminal or command prompt in the directory where the file is saved.
3. Compile the code using the command: **gcc calculator.c -o calculator -lm**
- 4 (Note: The **-lm** flag links the math library for `pow()` and `sqrt()`.)
- 5 4. Run the program using: **./calculator**

Example Output:

```
===== Advanced Calculator =====
1. Addition (+)
2. Subtraction (-)
3. Multiplication (*)
4. Division (/)
5. Power (x^y)
6. Square Root (√x)
7. Modulo (Remainder)
8. Exit
Choose an operation (1-8): 1
Enter two numbers: 10 5
Result: 15.00
```

Conclusion:

This project demonstrates structured programming in C using loops, conditional statements, and mathematical functions. It is a simple yet powerful example for beginners learning to apply core C concepts.